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HW5

1. $\mu = E[X]$ & $\sigma^2 = E[(X-\mu)^2]$ X is c.r.v. uniformly distributed on $[x_1, x_2] \subset \mathbb{R}$

so the pdf is defined as, $f_X(x) = \begin{cases} \frac{1}{x_2 - x_1} & \forall\ x_1 \leq x \leq x_2 \\ 0 & \text{otherwise} \end{cases}$

$$\mu = E[X] = \int_{x_1}^{x_2} x \cdot f_X(x)\,dx = \int_{x_1}^{x_2} \frac{x}{x_2 - x_1}\,dx$$

$$= \frac{1}{x_2 - x_1} \int_{x_1}^{x_2} x\,dx = \frac{1}{x_2 - x_1}\left[\frac{x^2}{2}\right]_{x_1}^{x_2} = \frac{1}{x_2 - x_1}\left(\frac{x_2^2 - x_1^2}{2}\right)$$

$$= \frac{x_1 + x_2}{2}$$

$Var(X) = E[X^2] - (E[X])^2 = \sigma^2$ (Need to prove)

$$E[X^2] = \int_{x_1}^{x_2} x^2 \cdot \frac{1}{x_2 - x_1} \cdot dx = \frac{1}{x_2 - x_1}\left[\frac{x^3}{3}\right]_{x_1}^{x_2}$$

$$= \frac{x_2^3 - x_1^3}{3(x_2 - x_1)} = \frac{(x_2 - x_1)(x_2^2 + x_1 x_2 + x_1^2)}{3(x_2 - x_1)}$$

$$= \frac{x_2^2 + x_1 x_2 + x_1^2}{3}$$

$$\therefore\ Var(X) = E[X^2] - (E[X])^2$$

$$= \frac{x_2^2 + x_1 x_2 + x_1^2}{3} - \left(\frac{(x_1 + x_2)}{2}\right)^2$$

$$= \frac{x_2^2 + x_1 x_2 + x_1^2}{3} - \frac{x_1^2 + 2x_1 x_2 + x_2^2}{4}$$

$$= \frac{4x_2^2 + 4x_1 x_2 + 4x_1^2 - 3x_1^2 - 6x_1 x_2 - 3x_2^2}{12}$$

$$= \frac{x_2^2 - 2x_1 x_2 + x_1^2}{12} = \frac{(x_2 - x_1)^2}{12}$$

Given $\sigma^2 = E[(X-\mu)^2]$

$$= E[X^2 - 2\mu X + \mu^2]$$

Using linearity of $E[\cdot]$ & $E[\text{constant}] = \text{constant}$ if constant $\in \mathbb{R}$

$$= E[X^2] - 2\mu \underbrace{E[X]}_{\mu} + \underbrace{E[\mu^2]}_{\mu^2}$$

$$= E[X^2] - 2\mu.\mu + \mu^2$$

$$= E[X^2] - \mu^2$$

or $= E[X^2] - (E[X])^2$ —— ① (Answer)

2 cdf of X,

$$F_X(x) = \begin{cases} 1 - 1/x, & \text{if } x \geq 1 \\ 0 & \text{if } x < 1 \end{cases}$$

we need to find pdf first which is for $x \geq 1$

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$= \frac{d}{dx}\left(1 - \frac{1}{x}\right) = 0 + \frac{1}{x^2} = \frac{1}{x^2}$$

for $x < 1$, $f_X(x) = 0$

$\therefore$ pdf is $= \begin{cases} 1/x^2 & x \geq 1 \\ 0 & x < 1 \end{cases}$

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

for $x \geq 1$ $E[X] = \int_1^{\infty} \frac{x}{x^2}\, dx = \int_1^{\infty} \frac{1}{x}\, dx = \lim_{b \to \infty} \int_0^b \frac{1}{x}\, dx$

$$= \lim_{b \to \infty} [\ln|x|]_1^b$$

$$= \lim_{\infty} [\ln(b) - \ln(1)] = \infty$$

Since $E[X] = \infty$, the expected value $E[X]$ does not exist

4 $Y = aX + b$ where $a, b \in \mathbb{R}$, $a \neq 0$

To find variance of Y in terms of X

from def^n $Var(Y) = E[(Y - E[Y])^2]$ — ①

$E[Y] = E[aX + b] = aE[X] + b$ — ②

② in ①

or $(Y - E[Y]) = ((aX + b) - (aE[X] + b))$

$= a(X - E[X])$

$\therefore Var(Y) = E[(a(X - E[X]))^2]$

$= E[a^2 (X - E[X])^2]$

$= a^2 E[(X - E[X])^2]$ → $Var(X)$

$Var(Y) = a^2 . Var(X)$ <u>Answer</u>

5 X r.v has mean μ_X and variance σ_X^2.

To find mean & variance of rv $Y = \frac{X - \mu}{\sigma}$

mean of r.v Y $E[Y] = E\left[\frac{X - \mu_X}{\sigma_X}\right]$

$= \frac{1}{\sigma_X} E[X - \mu_X]$

$= \frac{1}{\sigma_X}(E[X] - \mu_X)$

$= \frac{(\mu_X - \mu_X)}{\sigma_X} = 0$

$\therefore E[Y] = 0$

$$Var(Y) = Var\left(\frac{X - \mu_X}{\sigma_X}\right) \quad \Big| \quad Var(aX+b) = a^2 Var(X) \quad a, b = Const$$

$$= Var\left(\underbrace{\frac{X}{\sigma_X}}_{a} - \frac{\mu_X}{\sigma_X}\right)$$

$$\text{Thus } Var(Y) = \left(\frac{1}{\sigma_X}\right)^2 Var(X)$$

$$= \left(\frac{1}{\sigma_X}\right)^2 \cdot \sigma_X^2$$

$$= 1.$$

$$\therefore E[Y] = 0, \quad Var(Y) = 1. \quad \underline{\text{Ans}}$$

(6) Mean and Variance of binomial r.v

The prob that a r.v. X with binomial dist is

$$b(k, n, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

where k is sucess out of n independent trials when the prob of success in any one of trials is p.

$$\therefore E(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \sum_{k=0}^{n} k \cdot \frac{n!}{k!\,(n-k)!} p^k (1-p)^{n-k}$$

$$= \sum_{k=1}^{n} \frac{n!}{(k-1)!\,(n-k)!} p^k (1-p)^{n-k} = n \cdot p \sum_{1}^{n} \frac{(n-1)!}{(k-1)!\,(n-k)!} p^{k-1} (1-p)^{n-k} \quad \text{—(1)}$$

let $k-1 = y$ and $n-1 = x$ as at $k=0$, term vanish

$$\therefore \quad n p \sum_{y=0}^{x} \frac{x!}{y!\,(x-y)!} p^{y} (1-p)^{x-y}$$

(1) becomes

$$E[X] = n.p \sum_{y=0}^{x} \binom{x}{y} p^{y} (1-p)^{x-y} \quad \text{—— (2)}$$

from binomial theorem

$$(a+b)^m = \sum_{y=0}^{m} \frac{m!}{y!\,(m-y!} a^{y} b^{m-y}$$

here in (2) $a = p$, $b = 1-p$ & $m = x$

$$\therefore E[X] = n.p.(p + (1-p)^{x}$$

$$= np.$$

$$Var[X] = E[(X-\mu)^2]$$

$$= E[X^2] - [E[X]]^2$$

$E[X^2]$ $E[X(X-1)]$

let $k-2 = y$
& $n-2 = x$.
$k = y+2$
$y=0$.

$$= \sum_{k=0}^{n} k(k-1) \binom{n}{k} p^{k} (1-p)^{n-k}$$

$$= \sum_{k=0}^{n} k(k-1) \frac{n!}{(k)!\,(n-k)!} p^{k} (1-p)^{n-k}$$

$$= \sum_{k=2}^{n} \frac{n!}{(k-2)!\,(n-k)!} p^{k-2} (1-p)^{n-k}$$

$$= n(n-1).p^2 \sum_{y=0}^{x} \frac{x!}{y!\,(x-y)!} p^{y} (1-p)^{x-y}$$

$$= n(n-1)p^2 (p + (1-p))^{x}$$

$$= n(n-1)p^2$$

$$\therefore \operatorname{Var}(X) = E(X^2) - E(X)^2$$

$$= E(X(X-1)) + E(X) - E(X)^2$$

$$= n(n-1)p^2 + np - (np)^2 = np(1-p).$$

Ans

7 Mean and variance of the poisson random variable.

Let $X \sim \text{Poisson}(\lambda)$

The prob mass function is $P(X=k) = \dfrac{e^{-\lambda}\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$

The mean $E[X] = \sum_{k=0}^{\infty} k\, P(X=k)$

$$= \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{k\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{k\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}$$

$$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda \cdot \lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

$$\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = e^{\lambda}$$

$$= \lambda e^{-\lambda} \cdot e^{\lambda}$$

$$= \lambda$$

$$\operatorname{Var}(X) = E[X^2] - (E[X])^2 \quad \text{—— (1)}$$

Same as example before in prob 6 we compute

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\, P(X=k).$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{k(k-1)\lambda^k}{k!}$$

$$= \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \cdot e^{\lambda} \quad \text{since } \sum_{j=0}^{j=\infty} \frac{\lambda^j}{j!} = e^j$$

$$\therefore E[X^2] = E[X(X-1)] + \underbrace{E[X]}_{=\lambda^2} = \lambda^2 + \lambda$$

from ①, $Var(X) = (\lambda^2 + \lambda) - \lambda^2$

$$= \lambda$$

$$\left.\begin{array}{l} \therefore E[X] = \lambda \\ \& \ Var[X] = \lambda \end{array}\right\} \quad \underline{\text{Ans}}$$

⑧. Let X = number of people who get the correct hat

let $X_i = \begin{cases} 1 & \text{if } i^{th} \text{ person get their own hat} \\ 0 & \text{if not} \end{cases}$

$$\therefore X = X_1 + X_2 + \cdots + X_n$$

$$\therefore E[X] = E[X_1] + E[X_2] + \cdots + E[X_n]$$

(a) Prob that person i get his own hat $= \frac{1}{n}$ since equally likely,

$$\therefore E[X_i] = 1 \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right)$$

$$E[X_i] = \frac{1}{n}$$

$$\therefore E[X] = n \cdot \frac{1}{n} = 1.$$

$\therefore E[X] = 1$. or expected number of people who got their hat back is 1. <u>Ans</u>

5 (b) for $n \to \infty$ $\lim_{n\to\infty} E[X] = 1$ so the expected number of people who get their own hat is still 1.

Do we expect anyone to get their hat back → what we have n people with n hats

so all possible arrangement is $n!$

the arrangement where none gets their own hat

let that number of derangement be D_n

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} + \cdots + (-1)^n \frac{1}{n!}\right)$$

$$P(\text{none get correct hat}) = \frac{D_n}{n!}$$

$$= 1 - \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}$$

$$\lim_{x\to\infty} P(\text{none get their hat}) = e^{-1} = 0.368$$

so 36.8% of the time nobody gets their hat.

(9) $\sqcup_1 \sqcup_2$ $n \to \frac{1}{2}$

if we place a ball it goes into any bag with prob $\frac{1}{n}$.

for specific bag i, prob of success $p = \frac{1}{n}$ where success

(a) means ball goes inside bag i

the number of balls needed until the first success follow geometric distribution

$$E[X] = \frac{1}{p}$$

$$\therefore E[X] = 1/\tfrac{1}{n} = n$$

Ans

8 (b) if $n \to \infty$ $\lim_{n\to\infty} E[X] = 1$. so the expected number of people who get their own hat is still 1.

Do we expect anyone to get their hat back →

what we have n people with n hats

so all possible arrangement is $n!$

the arrangement where noone gets their own hat

let that number of derangement be D_n

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} + \cdots + (-1)^n \frac{1}{n!}\right)$$

$$P(\text{noone get correct hat}) = \frac{D_n}{n!}$$

$$= 1 - \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}$$

$$\lim_{x\to\infty} P(\text{noone get their hat}) = e^{-1} = 0.368$$

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Ans

9
(b) To show the expected number of balls one need to put in before every bag has at least one ball is ~~n (H)(n)~~ $n\,H(n)$ where $H(n) = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$

Let $T \to$ total number of balls placed until every bag has at least one ball (inclusive)

let $t_k \to$ number of additional balls needed to obtain k th new bag.

$$T = \sum_{k=1}^{n} t_k$$

when $k-1$ bags already contain balls, empty bags $= n-(k-1)$

prob the next ball hits a new bag:

$$p_k = \frac{n-k+1}{n}$$

each t_k follows geometric distribution with success prob p_k $\therefore$ $E[t_k] = \frac{1}{p_k} = \frac{n}{n-k+1}$

$$E[T] = \sum_{k=1}^{n} E[T_k] = \sum_{k=1}^{n} \frac{n}{n-k+1}$$

$$= n \sum_{m=1}^{n} \frac{1}{m} \quad \text{where } m = n-k+1$$

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

$$= n\,H_n$$

Ans

5 prob of heads $= p$, tails $1-p$

it is flipped until either heads / tails have occurred twice, we want the expected number of flips until either heads or tails occurs twice let call it E.

let E_H = expected flips remaining if first flip is H

E_T = " " " " " " T

$\therefore E = 1 + p.E_H + (1-p)\, E_T$ so example HH or TT

$E_H = p(2) + (1-p)(3) = 3-p$ — (1)

if first flip is H:

second flip H we stop so flips are 2 & if second flip T, we stop on ext flip (third one) flips are 3.

example HTH, HTT, THH, THT

$P(HH) = p^2$

$P(TT) = (1-p^2)$

$P(HTH) = p(1-p)p = p^2(1-p)$

$P(HTT) = p(1-p)(1-p) = p(1-p)^2$

$P(THH) = (1-p)p^2$ & $P(THT) = (1-p)^2 . p$

$\therefore E_T = (1-p).2. + p(3) = 2(1-p) + 3p = 2+p$ — (2)

$\therefore E = 1 + p\,E_H + (1-p)\,E_T = 1 + p(3-p) + (1-p)(2+p)$

or ~~$E = 2(1-p) + p(1-p)\ 2-p-p^2$~~

$\therefore E = 1 + 3p - p^2 + 2 - p - p^2$

$= 3 + 2p - 2p^2$

Ans